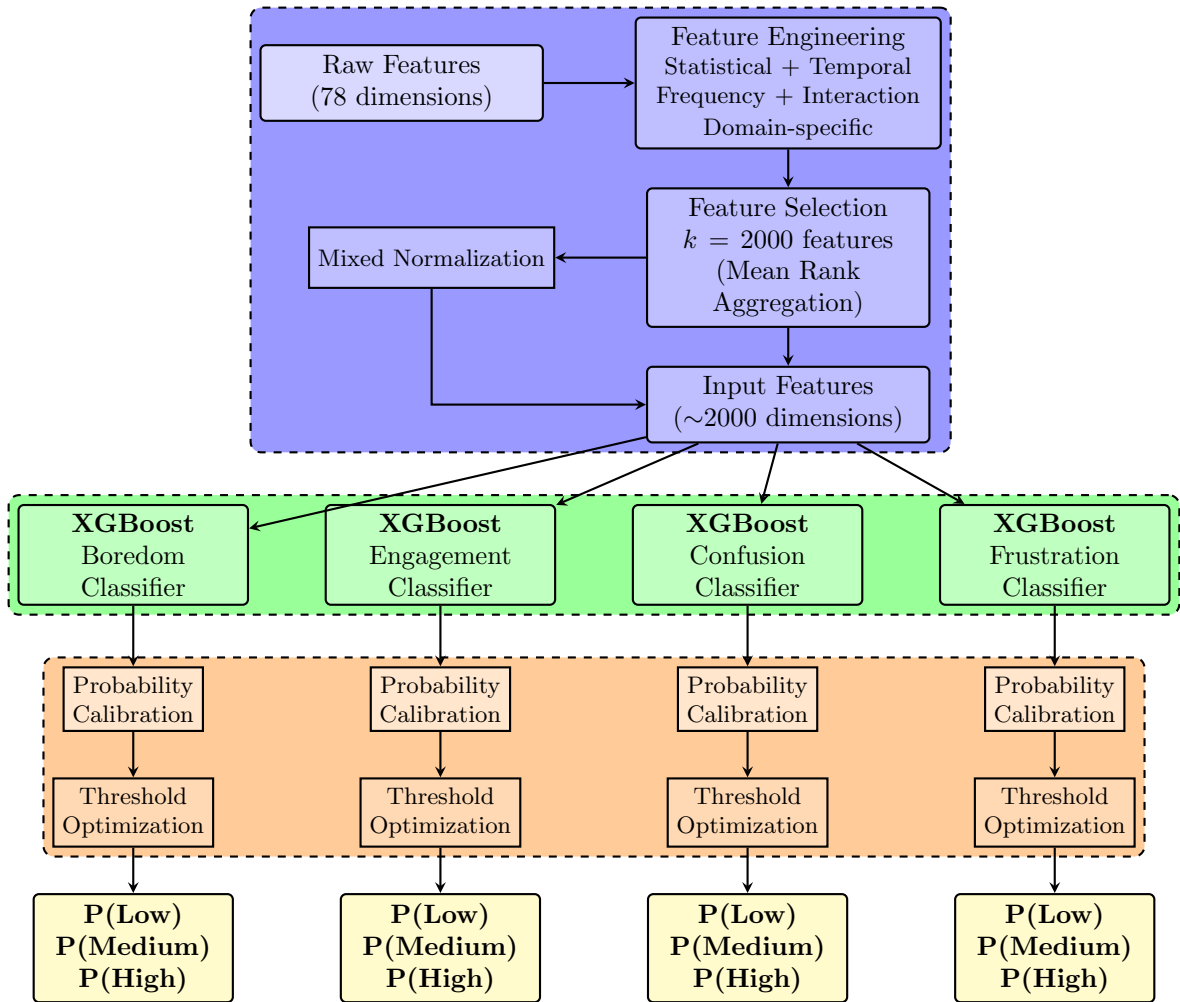




XGBoost Hyperparameters:

- max_depth=8
- learning_rate=0.05
- n_estimators=500
- subsample=0.8
- colsample_bytree=0.8
- min_child_weight=3
- gamma=0.1
- reg_alpha=0.1 (L1)
- reg_lambda=1.0 (L2)
- objective=multi:softprob
- early_stopping_rounds=50

Class Imbalance Handling:

- SMOTE Oversampling: $k=5$ neighbors
- Class Weights: Inverse frequency
- Focal Loss: $\alpha=0.25$, $\gamma=2.0$

Calibration & Thresholding:

- Calibration Method: Isotonic regression
- Threshold Metric: F1-macro
- Search Range: $[0.2, 0.8]$